# TEJASHVI RAJ

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## EDUCATION

**Vellore Institute of Technology** — 2022 – 2026
*Bachelor of Technology - Electronics and Communication Engineering; CGPA: 8.63/10* — Bhopal, Madhya Pradesh

## SKILLS

**Programming Languages** : C++, Embedded C, Python
**Tools and Platforms** : ESP32, Arduino, Raspberry Pi, KiCad, Blynk IoT, LTSpice, Tinkercad, SIMULINK, Github
**Domains**: Embedded Systems, VLSI Design, IoT, Electronics Hardware, PCB Design, Sensor Interfacing and Calibration

## PROJECTS

**DHARA : The Ultimate Aqua-Hydro Manager** | **View Project**
*Embedded Systems, IoT*

- Developed an ESP32-based real-time monitoring system integrating TDS, pH, turbidity, and temperature sensors for hydroponic/aquaponic applications with automated pH regulation and cloud analytics
- **Technologies**: ESP32, Arduino Uno, Sensors, Ubidots IoT, C++
- **Role**: Embedded logic design, system integration, sensor calibration, and UART-based communication between microcontrollers for real-time data exchange

**Smart Bin: IoT-Based Touchless Dustbin** | **View Project**
*Embedded Systems, IoT*

- Designed a smart waste monitoring system using ESP32 and ultrasonic sensing with less than 10 ms alert response, enabling real-time overflow detection and cloud notifications
- **Technologies**: ESP32, Ultrasonic Sensor, Blynk IoT, C++
- **Role**: Sensor integration, calibration, real-time system monitoring, and firmware optimization for efficient performance

**FSM-Based Counter Design Automation Tool** | **View Tool**
*VLSI / Digital Design Automation*

- Built an automated FSM-based framework to synthesize synchronous counters from user-defined state sequences with minimized flip-flop excitation equations, improving design efficiency
- **Technologies**: Verilog HDL, FSMs, Digital Logic Design, Python, SymPy, Flask, HTML/CSS
- **Role**: FSM modeling, Boolean minimization, HDL generation, and development of a web-based automation tool

## EXPERIENCE

**Project Intern** — Nov. 2025 – Dec. 2025
*Shitashii Innovations Private Limited* — IIT Kanpur, UP

- Worked on PCB design and schematic development using KiCad
- Assisted in Raspberry Pi-based hardware interfacing and embedded system tasks
- Contributed to circuit layout optimization and component-level debugging

## CERTIFICATIONS

- **Embedded System Design Certification** (Maven Silicon, 2025) — Link
- **VLSI Design Internship – RISC-V and RTL Fundamentals** (Maven Silicon, 2025) — Link
- **MongoDB Associate Database Administrator** (FacePrep, 2025) — Link
- **PCB Design Using KiCad** (Pantech AI, 2025) — Link

## CO-CURRICULAR

**Co-Head, Electronics Team, VITroniX Club** — Jul. 2024 – Feb. 2025